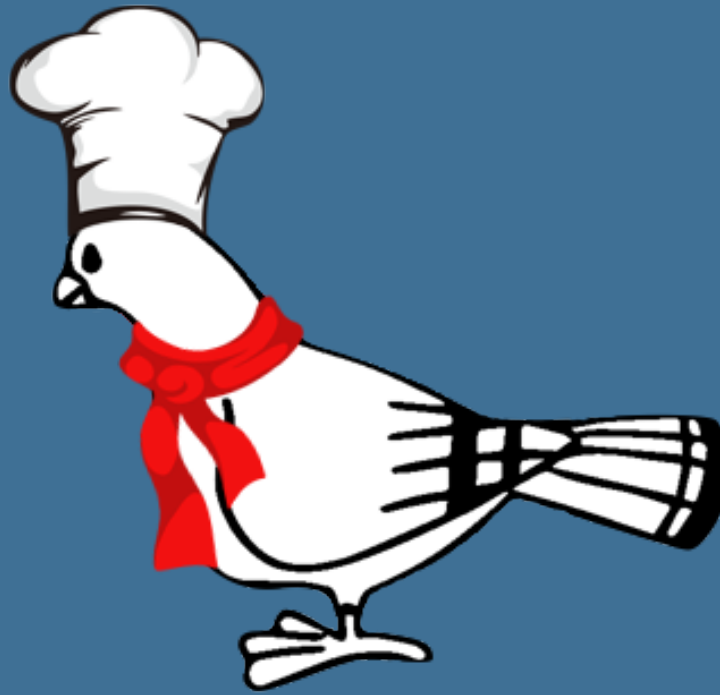




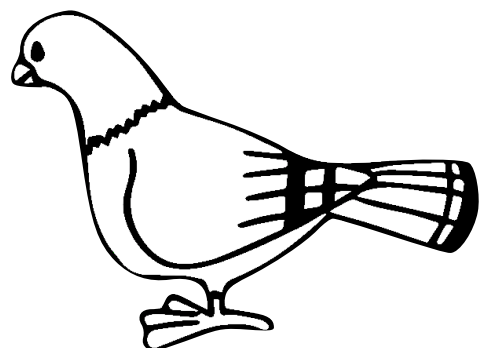
PLUMED Masterclass

21.3: Umbrella sampling

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 @BussiGio





PLUMED

open-source
freely-available
C++ library

- enhanced-sampling methods
- free-energy methods
- analysis MD data



www.plumed.org



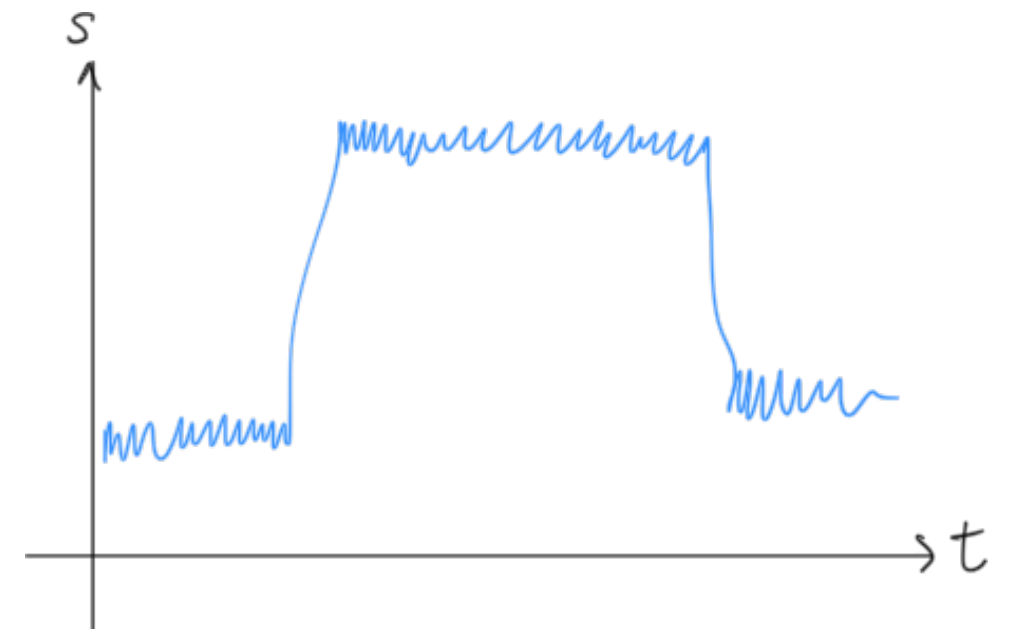
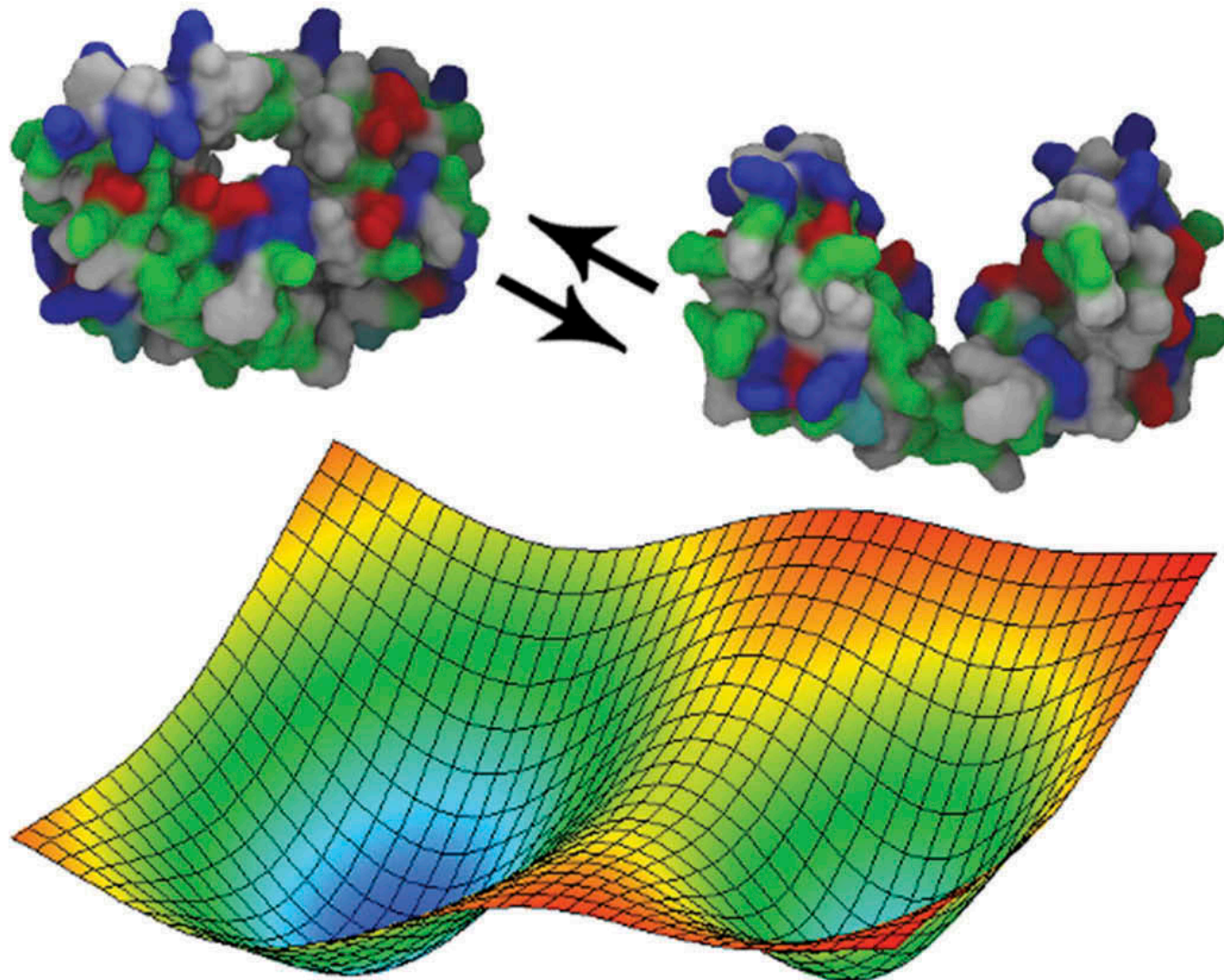
@plumed_org

Bonomi *et al.* CPC 2009
Tribello *et al.* CPC 2014

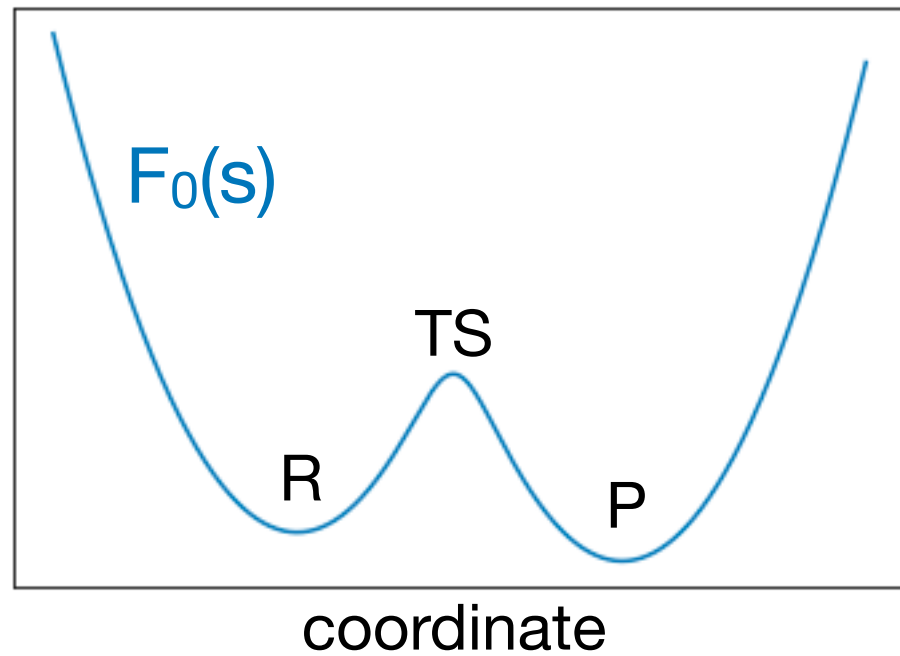


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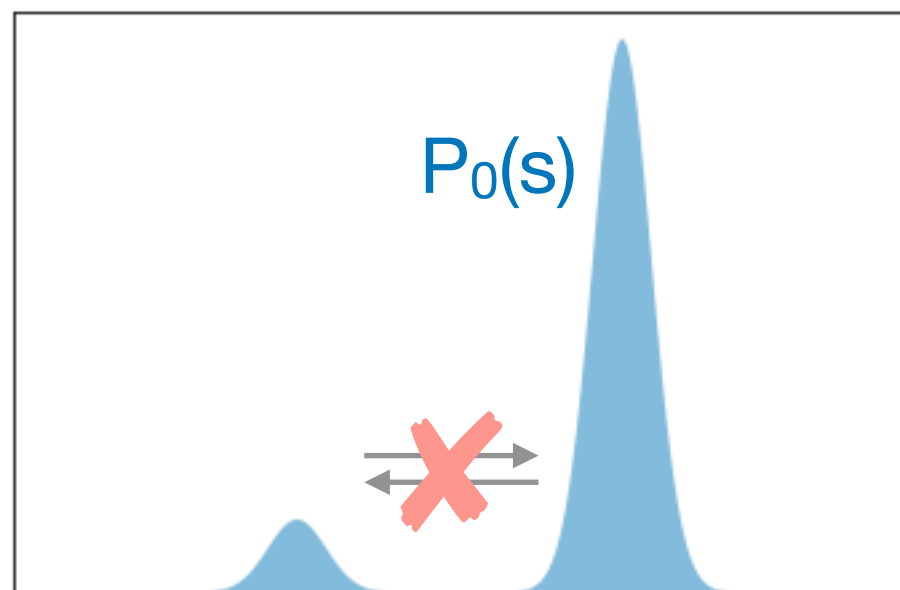
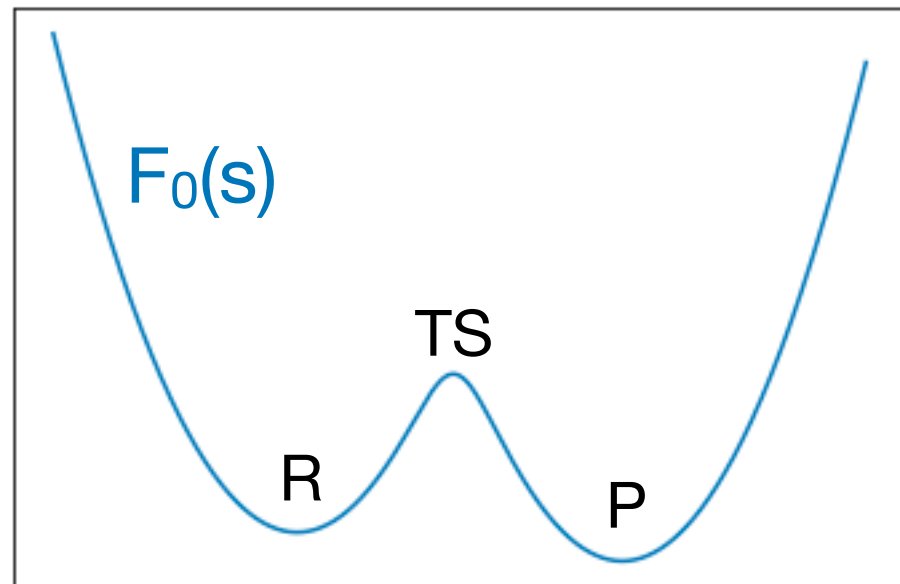
Rare events



How biased sampling could help: umbrella sampling

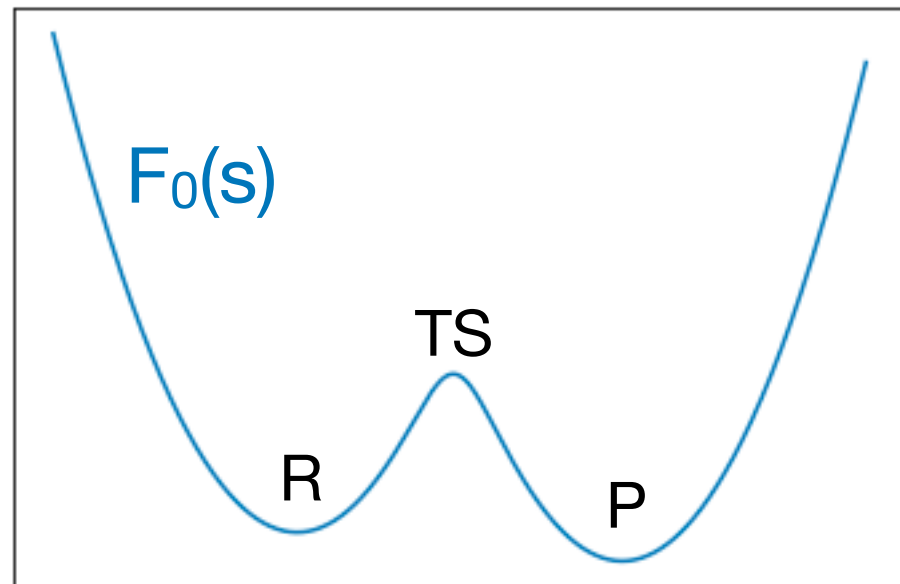


How biased sampling could help: umbrella sampling

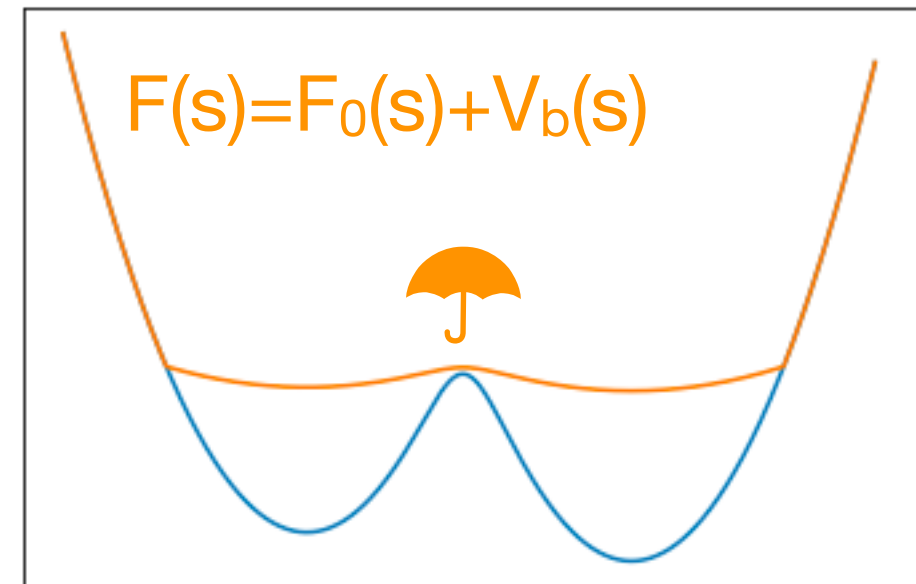


coordinate

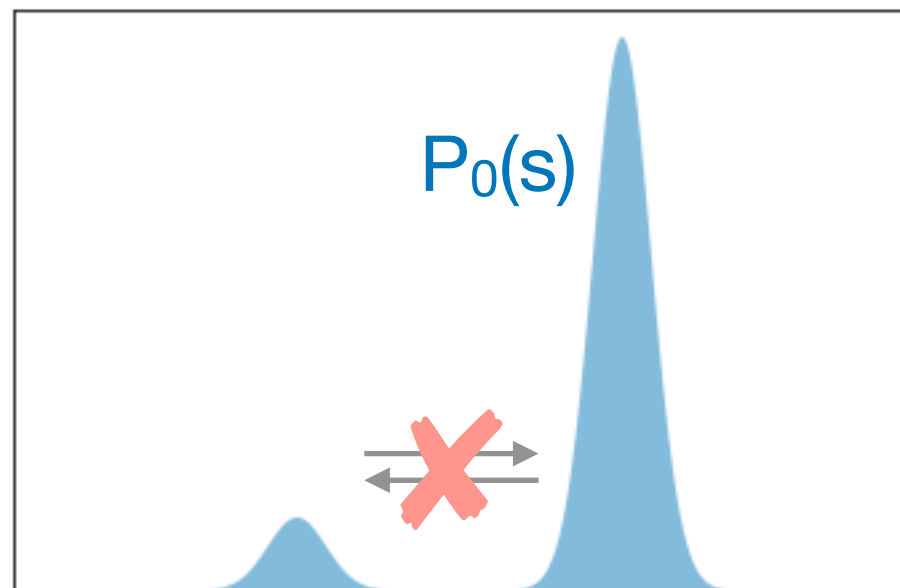
How biased sampling could help: umbrella sampling



coordinate

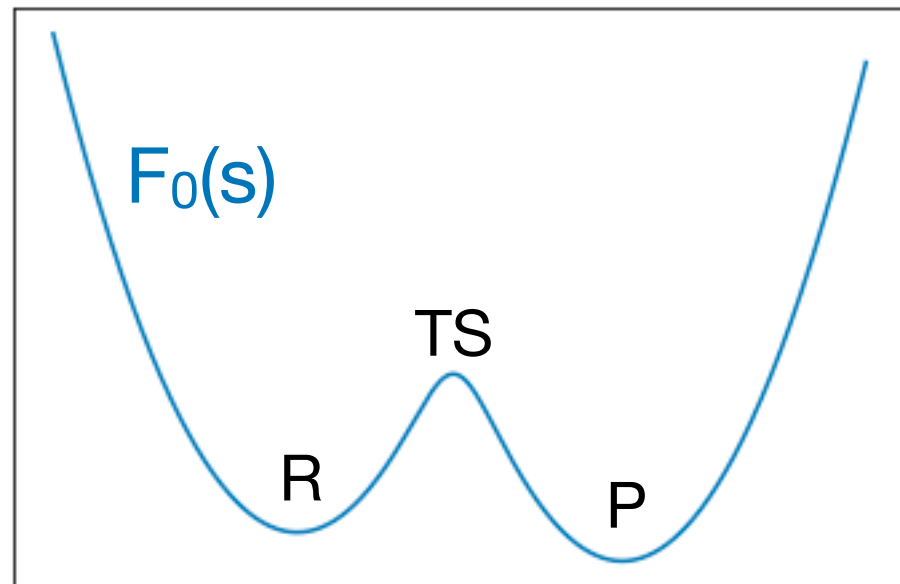


coordinate

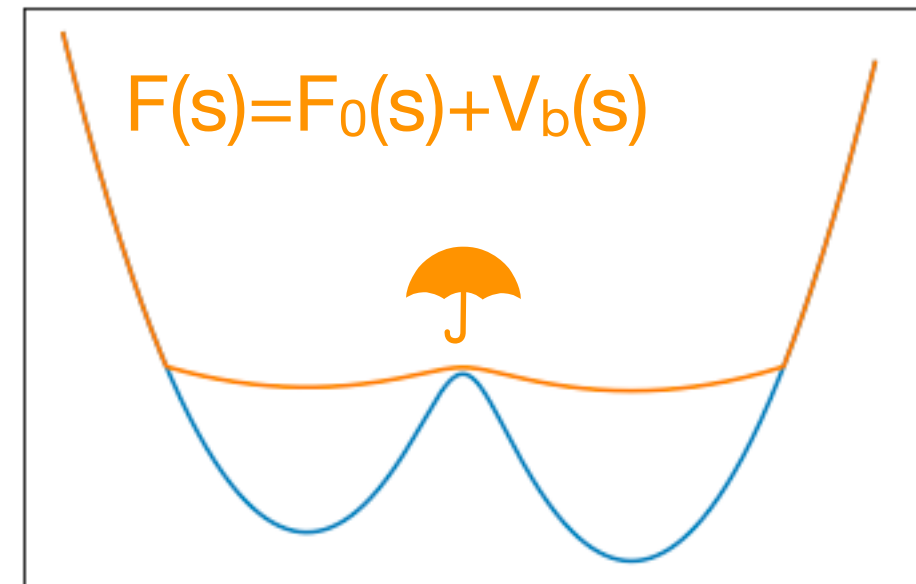


coordinate

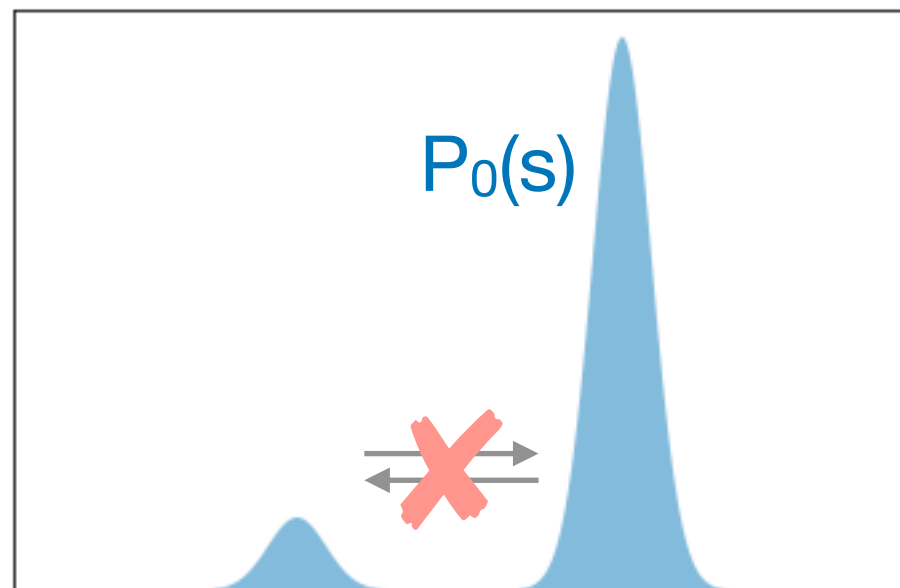
How biased sampling could help: umbrella sampling



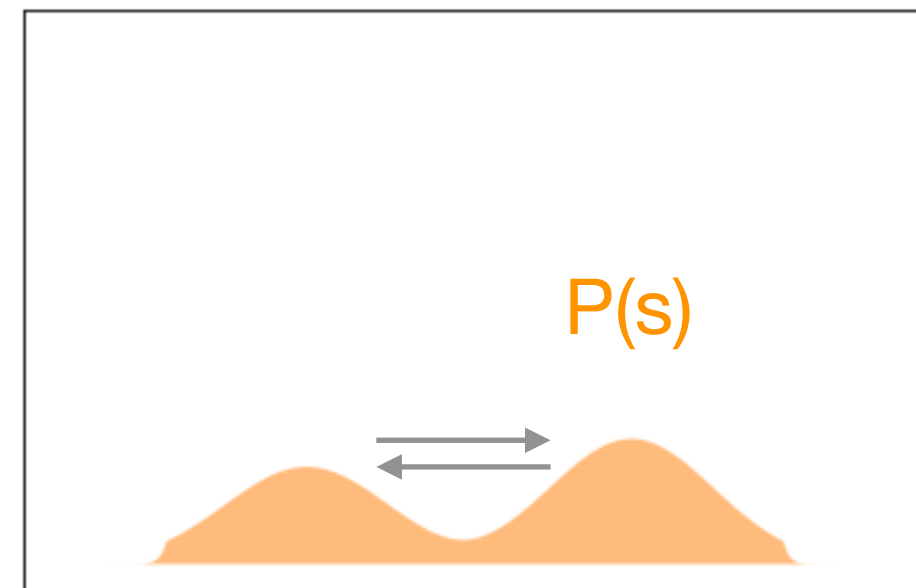
coordinate



coordinate

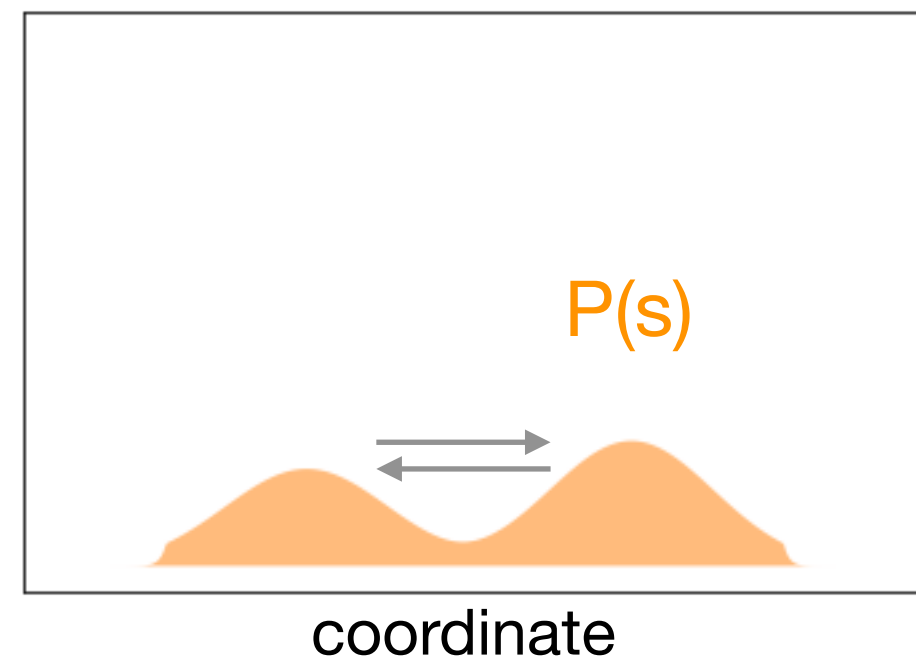
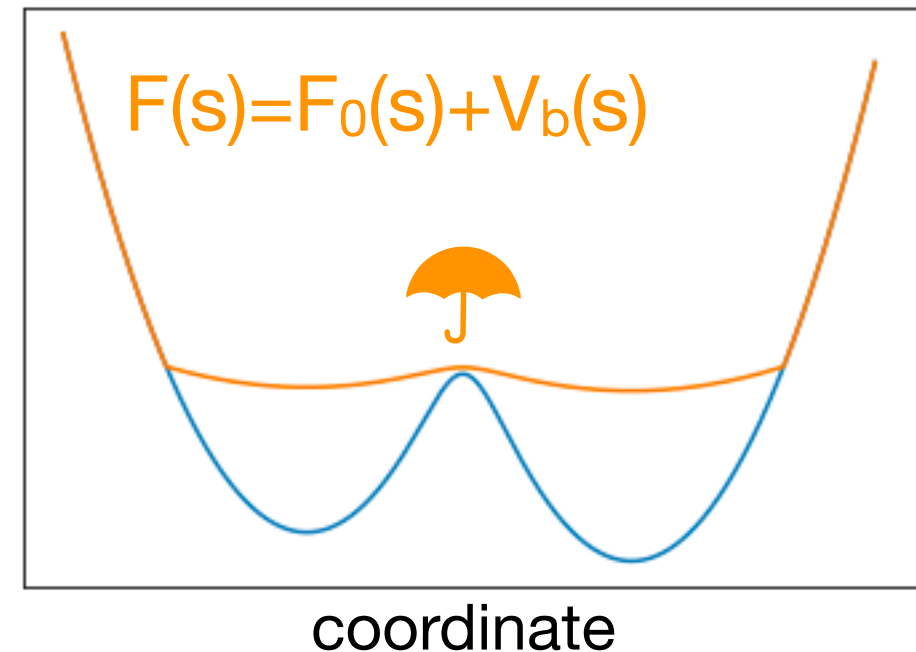
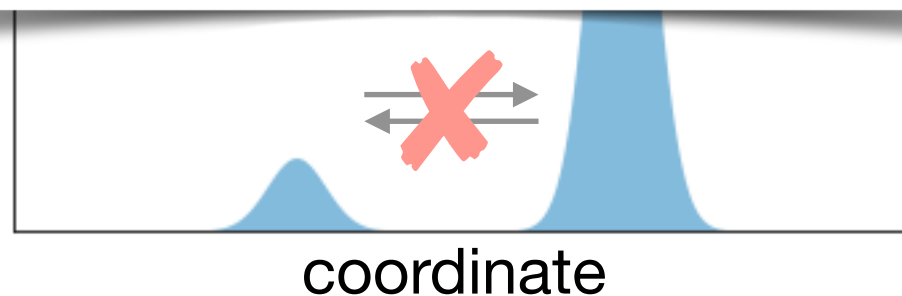
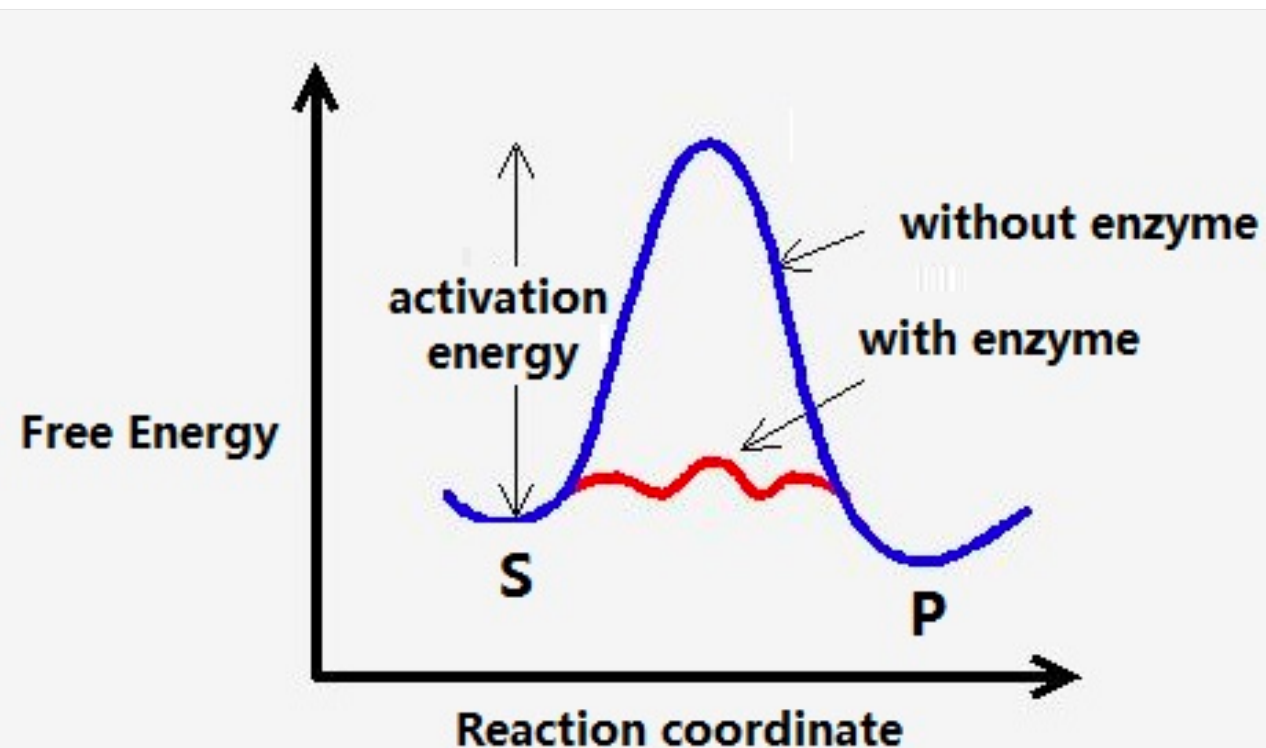


coordinate



coordinate

How biased sampling could help: umbrella sampling



Disfavor reactant and product (i.e. favor transition state), like a biological enzyme

Unbiasing the result

how much the bias V_b affects the probability

number of times state j was visited

$$P(t_1, \dots, t_M) \propto \prod_{j=1}^M \left(e^{-\beta V_b(s_j)} p_j \right)^{t_j}$$

product over all states

unbiased probability of observing state j

Unbiasing the result

how much the bias V_b affects the probability

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$$P(t_1, \dots, t_M) \propto \prod_{j=1}^M \left(e^{-\beta V_b(s_j)} p_j \right)^{t_j}$$

product over all states

unbiased probability of observing state j

Find p that maximises the likelihood

$$w(s) \propto e^{+\beta V_b(s)}$$

Averages can be computed by reweighing

$$\langle A \rangle = \frac{\sum_j A_j w(s_j)}{\sum_j w(s_j)}$$

Unbiasing the result

how much the bias V_b affects the probability

number of times state j was visited

$$P(t_1, \dots, t_M) \propto \prod_{j=1}^M \left(e^{-\beta V_b(s_j)} p_j \right)^{t_j}$$

product over all states

unbiased probability of observing state j

Find p that maximises the likelihood

Weight more if system went there even if disfavored

$$w(s) \propto e^{+\beta V_b(s)}$$

Averages can be computed by reweighing

$$\langle A \rangle = \frac{\sum_j A_j w(s_j)}{\sum_j w(s_j)}$$

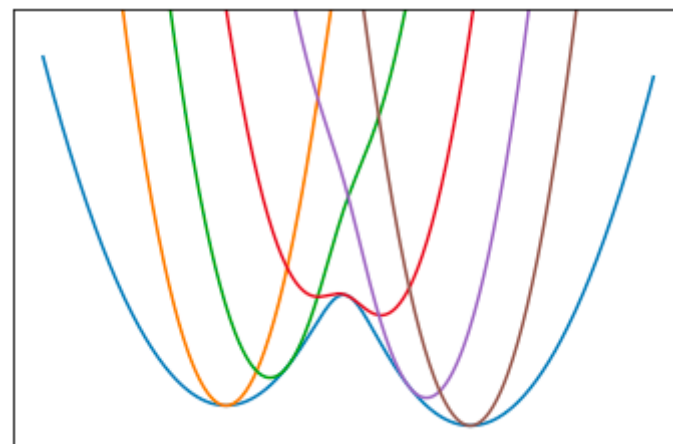
Multiple-windows umbrella sampling

Each simulation estimates the profile for a narrow range of the biased variable

stiffness

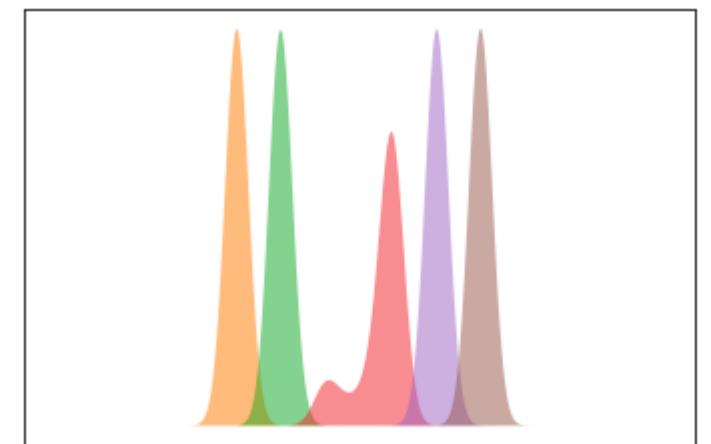
$$V_b^j(s) = \frac{k}{2} (s - s_j)^2$$

interpolate
between
reactant and
product



coordinate

Overlapping
histograms



coordinate

Combining multiple simulations: WHAM

how much the k -th bias affects the probability

number of times state j was visited in traj k

$$P(t_1^1, \dots, t_M^N) \propto \prod_{j=1}^M \prod_{k=1}^N \left(c_k e^{-\beta V_b^k(s_j)} p_j \right)^{t_{kj}}$$

product over all states and all trajectories

make sure each probability is normalized

unbiased probability of observing state j

Combining multiple simulations: WHAM

$$P(t_1^1, \dots, t_M^N) \propto \prod_{j=1}^M \prod_{k=1}^N \left(c_k e^{-\beta V_b^k(s_j)} p_j \right)^{t_{kj}}$$

how much the k -th bias affects the probability $\rightarrow$ $c_k e^{-\beta V_b^k(s_j)}$
 number of times state j was visited in traj k $\rightarrow$ t_{kj}
 unbiased probability of observing state j $\rightarrow$ p_j
 product over all states and all trajectories $\rightarrow$ $\prod_{j=1}^M \prod_{k=1}^N$
 make sure each probability is normalized $\rightarrow$ c_k

Find p that maximises the likelihood

Combining multiple simulations: WHAM

$$P(t_1^1, \dots, t_M^N) \propto \prod_{j=1}^M \prod_{k=1}^N \left(c_k e^{-\beta V_b^k(s_j)} p_j \right)^{t_{kj}}$$

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 product over all states and all trajectories $\rightarrow$ $\prod_{j=1}^M \prod_{k=1}^N$
 make sure each probability is normalized $\rightarrow$ c_k

Find p that maximises the likelihood

$$c_k = \frac{1}{\sum_j e^{-\beta V_b^k(s_j)} w(s_j)}$$

$$w(s) \propto \frac{1}{\sum_k c_k e^{-\beta V_b^k(s)}}$$

Self-consistent equations

Combining multiple simulations: WHAM

$$P(t_1^1, \dots, t_M^N) \propto \prod_{j=1}^M \prod_{k=1}^N \left(c_k e^{-\beta V_b^k(s_j)} p_j \right)^{t_{kj}}$$

how much the k -th bias affects the probability $\rightarrow$ $c_k e^{-\beta V_b^k(s_j)}$
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 make sure each probability is normalized $\rightarrow$ c_k
 product over all states and all trajectories $\rightarrow$ $\prod_{j=1}^M \prod_{k=1}^N$

Find p that maximises the likelihood

Averages can be computed by reweighing

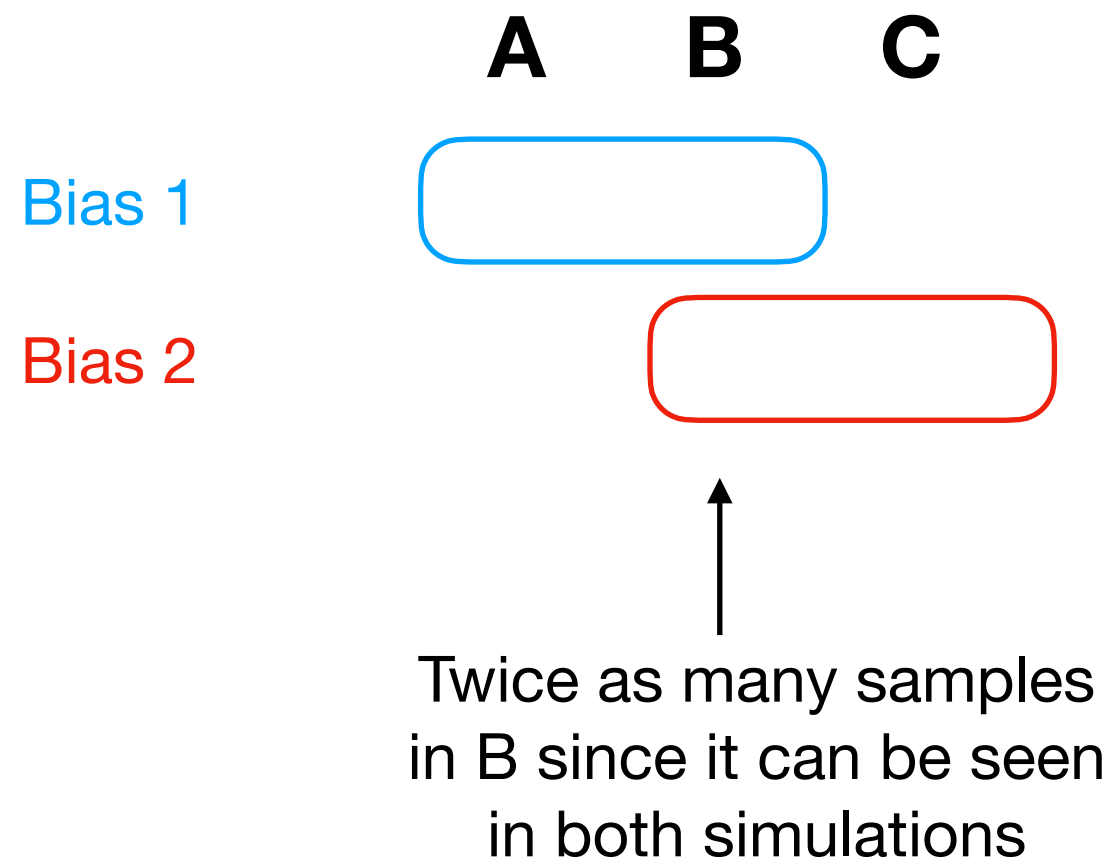
$$c_k = \frac{1}{\sum_j e^{-\beta V_b^k(s_j)} w(s_j)}$$

$$w(s) \propto \frac{1}{\sum_k c_k e^{-\beta V_b^k(s)}}$$

Self-consistent equations

$$\langle A \rangle = \frac{\sum_j A_j w(s_j)}{\sum_j w(s_j)}$$

Combining multiple simulations: WHAM



$$c_k = \frac{1}{\sum_j e^{-\beta V_b^k(s_j)} w(s_j)}$$

$$w(s) \propto \frac{1}{\sum_k c_k e^{-\beta V_b^k(s)}}$$

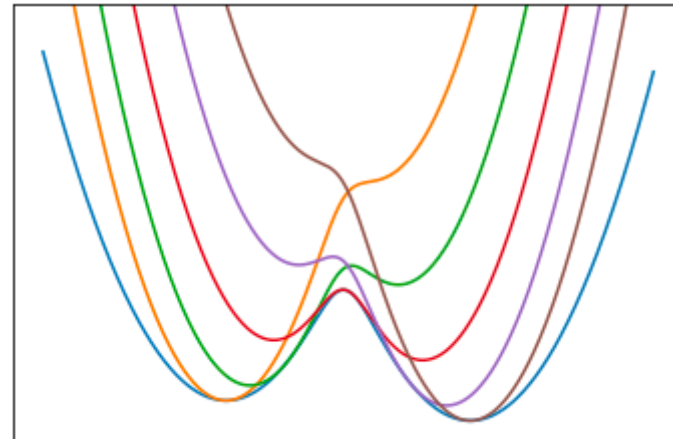
Self-consistent equations

Pay attention to stiffness

stiffness

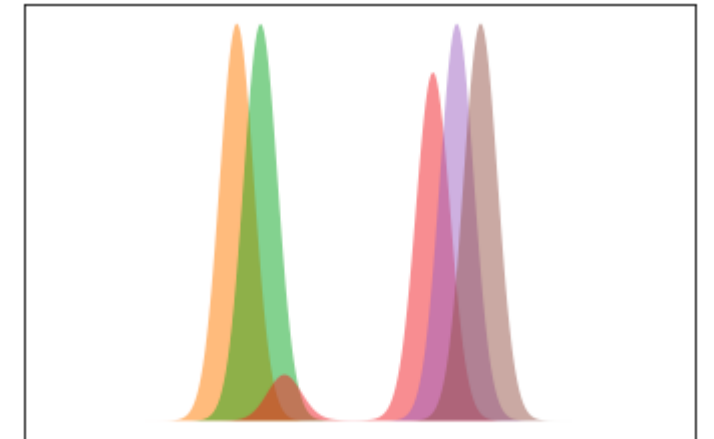
$$V_b^j(s) = \frac{k}{2} (s - s_j)^2$$

interpolate
between
reactant and
product

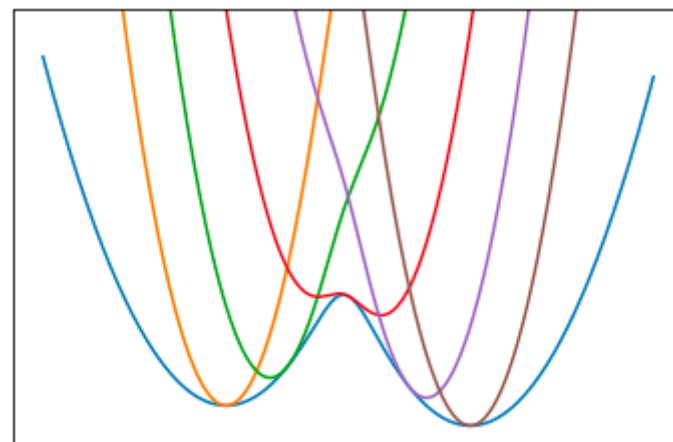


coordinate

Too soft:
Not sampling
TS

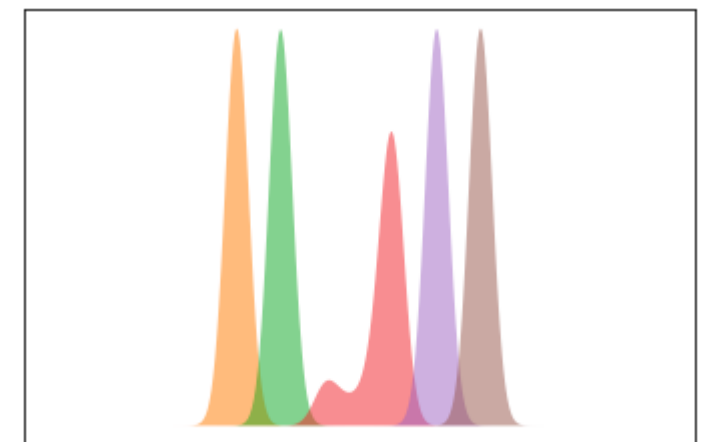


coordinate

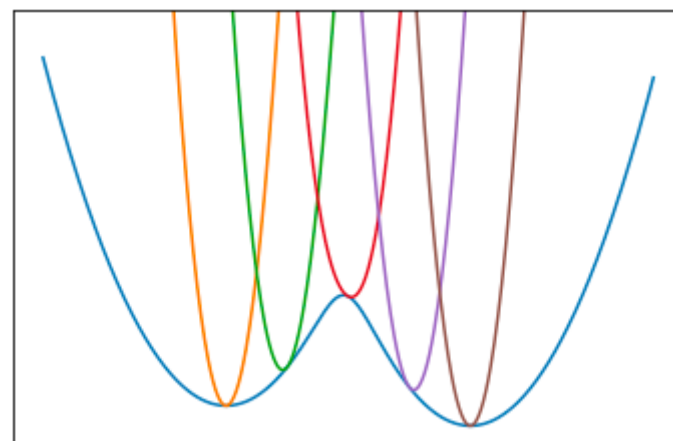


coordinate

Overlapping
histograms

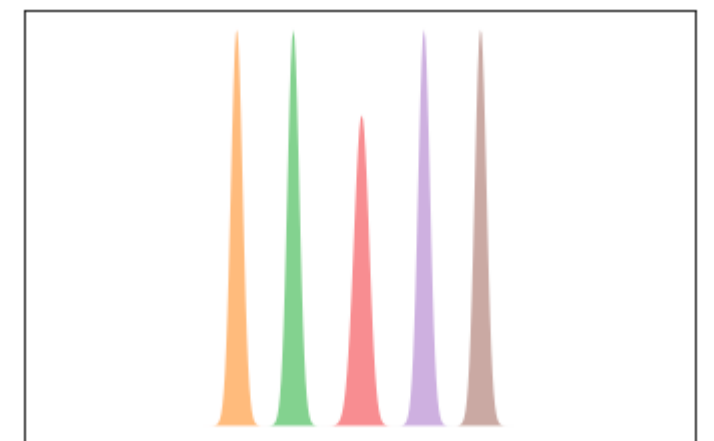


coordinate



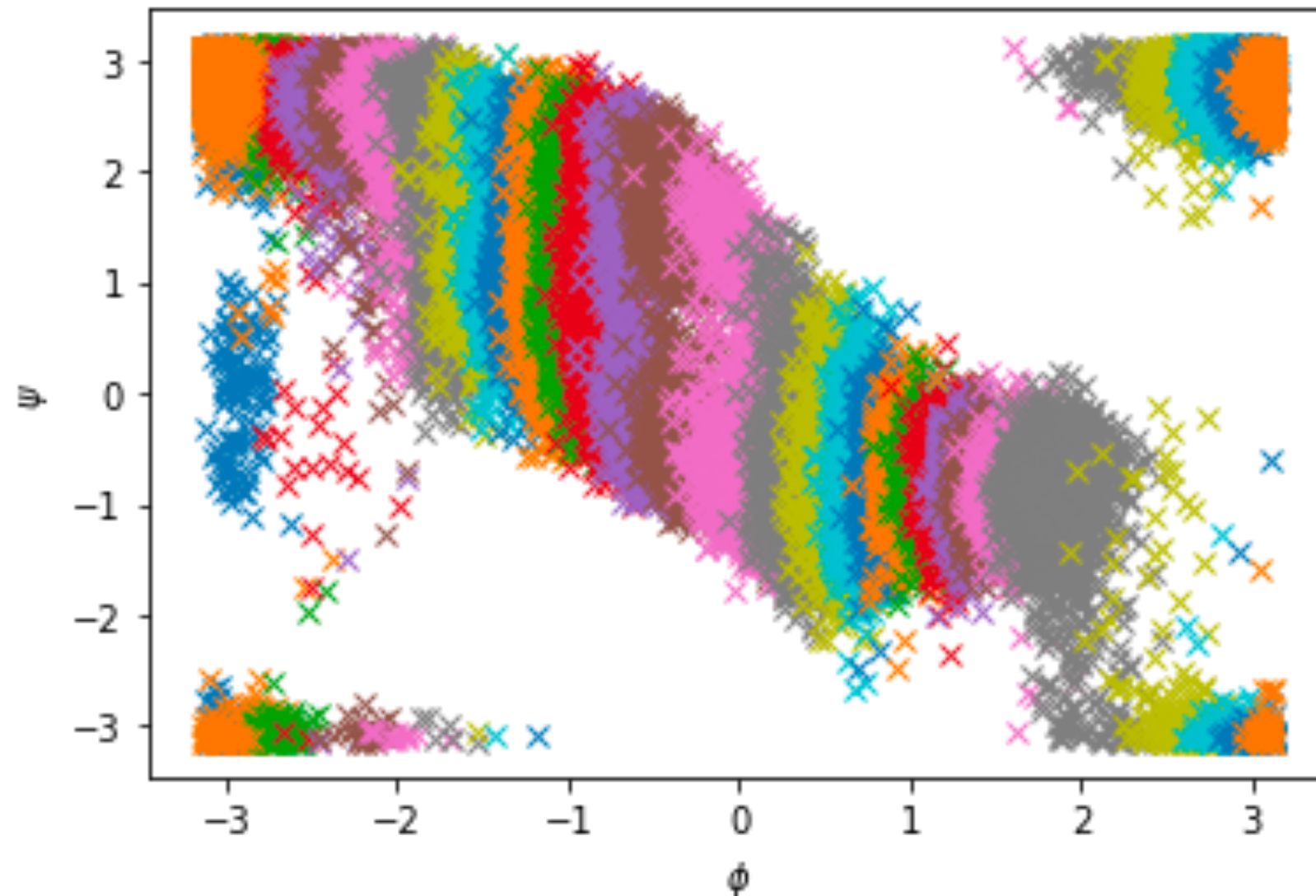
coordinate

Too stiff:
No overlap!



coordinate

Checking for overlap

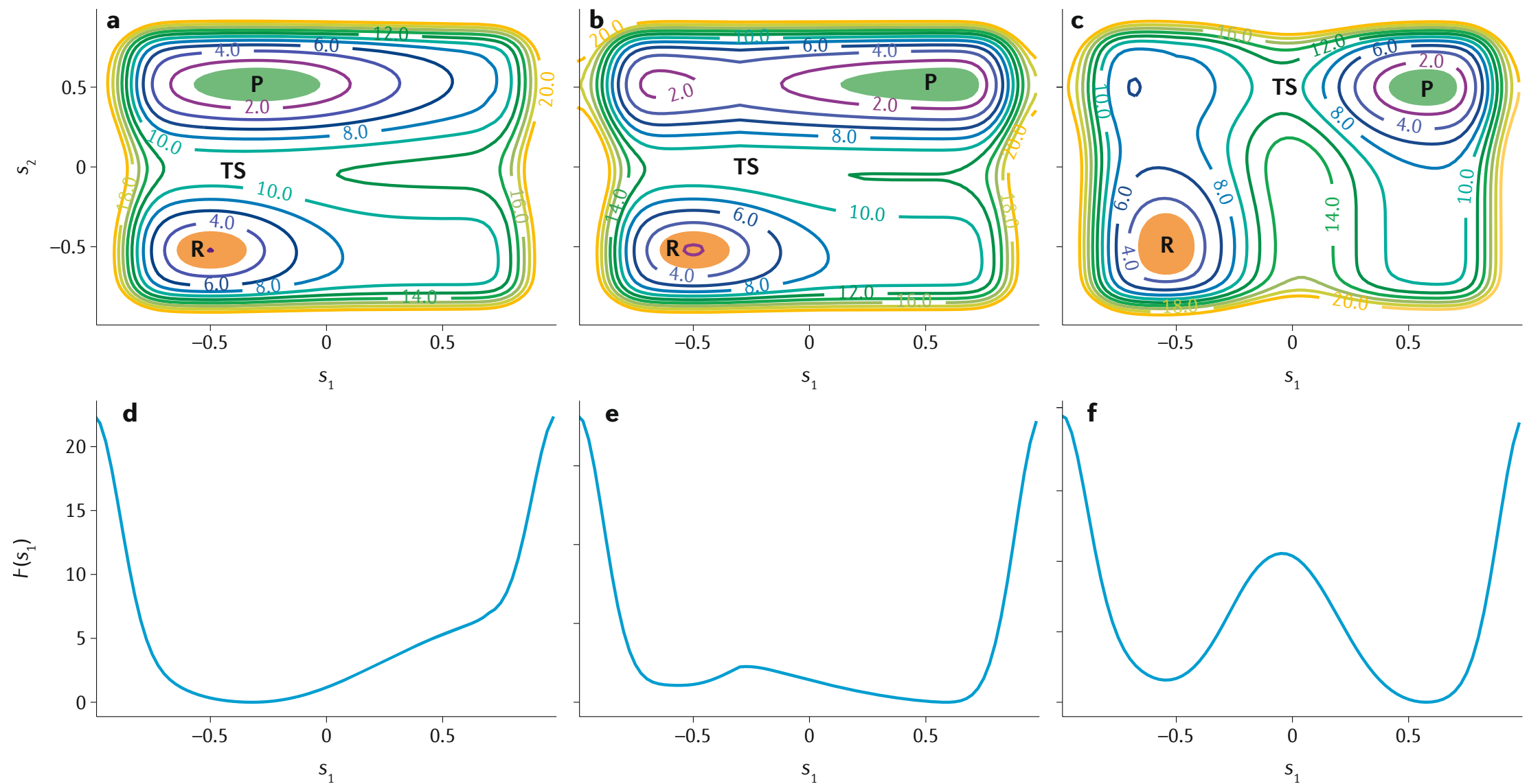


Simulations should overlap also in not-biased degrees of freedom

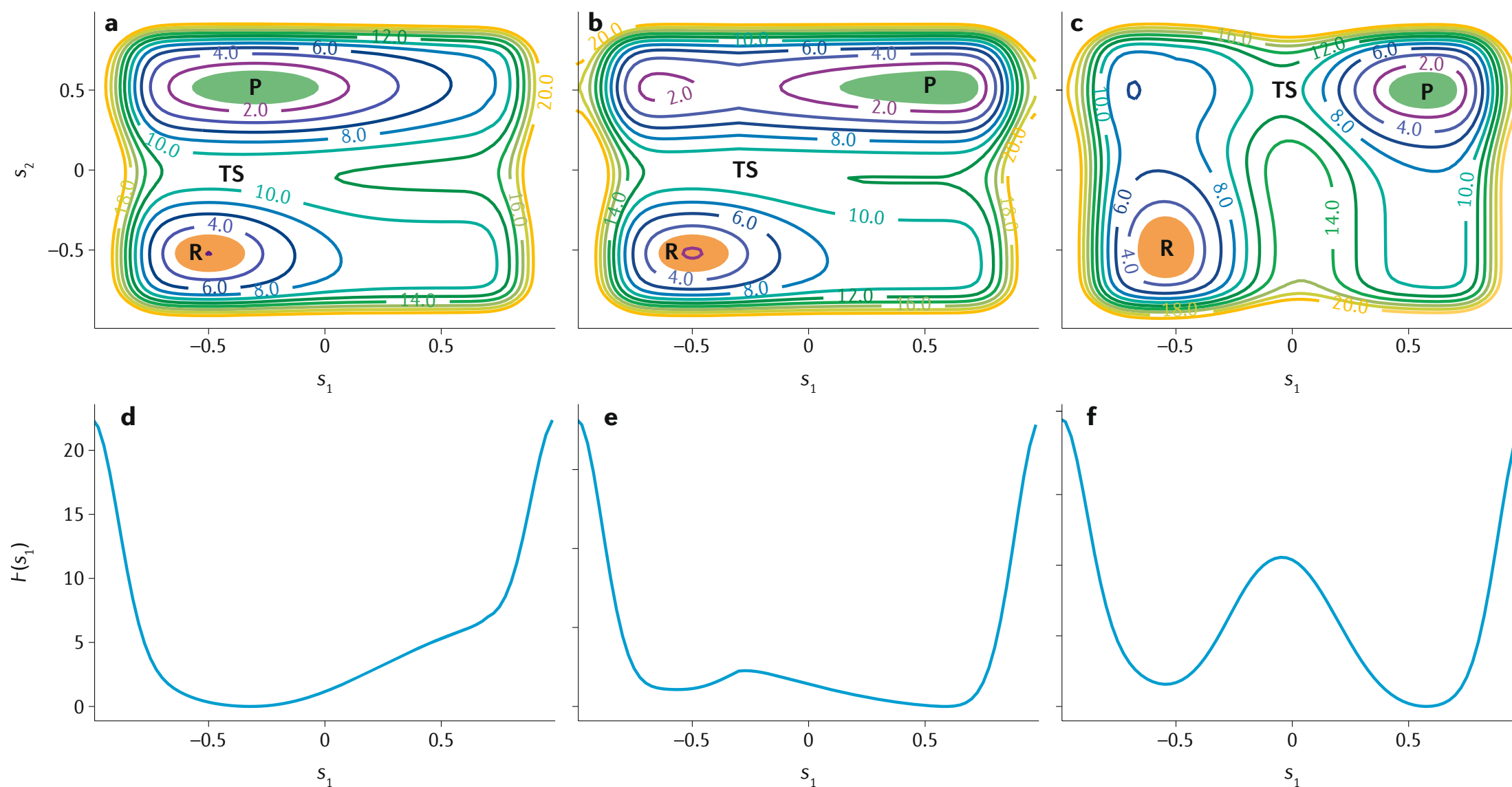
Very difficult to check without doing replica exchange (come to masterclass 4!)

A minimal check: compare results initialising in R vs P

Reaction coordinates: good ones vs bad ones



Reaction coordinates: good ones vs bad ones



	$R \approx P$	$R \approx TS \neq P$ or vice versa	$R \neq TS \neq P$
Analysis	😓	😊	😊
Enhanced sampling	😓	😓	😊

Error calculations



time series



analysis

A

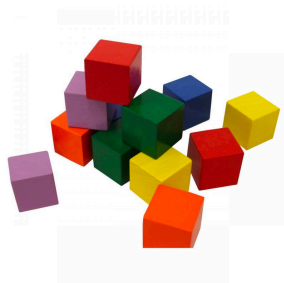
Error calculations



time series

analysis

A



blocks - minimize correlations

0 1 2 3 4 5 6 7 8 9

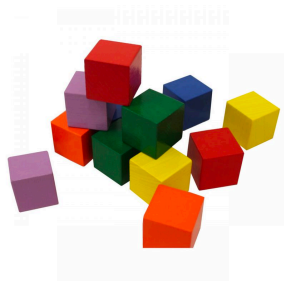
Error calculations



time series

analysis

A



blocks - minimize correlations

0	1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---	---



bootstrap samples

0	5	0	5	2	8	8	4	9	0	1
1	1	9	8	1	9	5	9	4	6	1
2	7	2	4	2	3	9	4	8	8	4
...	...	...	...	...	...	...	...	...	...	...
198	5	2	3	2	4	5	0	6	8	5
199	6	5	6	6	2	5	3	2	5	2

Error calculations

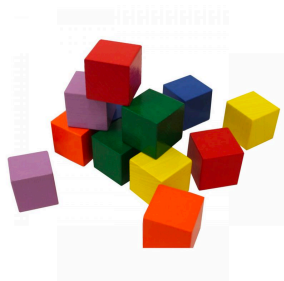


time series



analysis

A



blocks - minimize correlations

0	1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---	---



bootstrap samples

0	5	0	5	2	8	8	4	9	0	1
1	1	9	8	1	9	5	9	4	6	1
2	7	2	4	2	3	9	4	8	8	4
...	...	...	...	...	...	...	...	...	...	...
198	5	2	3	2	4	5	0	6	8	5
199	6	5	6	6	2	5	3	2	5	2

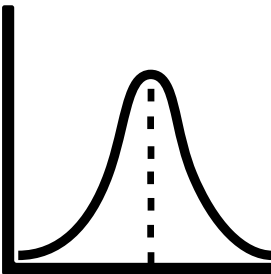
A₀

A₁

A₂

A₁₉₈

A₁₉₉



Useful references on WHAM

Ferrenberg and Swendsen, PRL (1989)
Initial formulation

Kumar et al JCC (1992)
First example in biomolecular simulations

Souaille and Roux CPC (2001)
Weights for individual frames

Shirts and Chodera JCP (2008)
MBAR, can be shown to be equivalent to binless WHAM

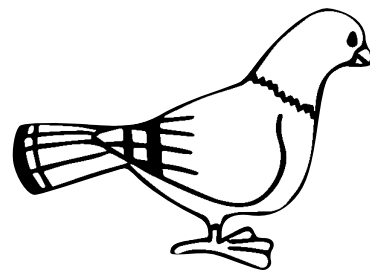
Hub et al, JCTC (2010)
Estimating error using bootstrap

Tan, Gallicchio, Lapelosa, and Levy JCP (2012)
Show that WHAM can be used in a binless fashion

Using PLUMED for on-the-fly analysis

MD engine

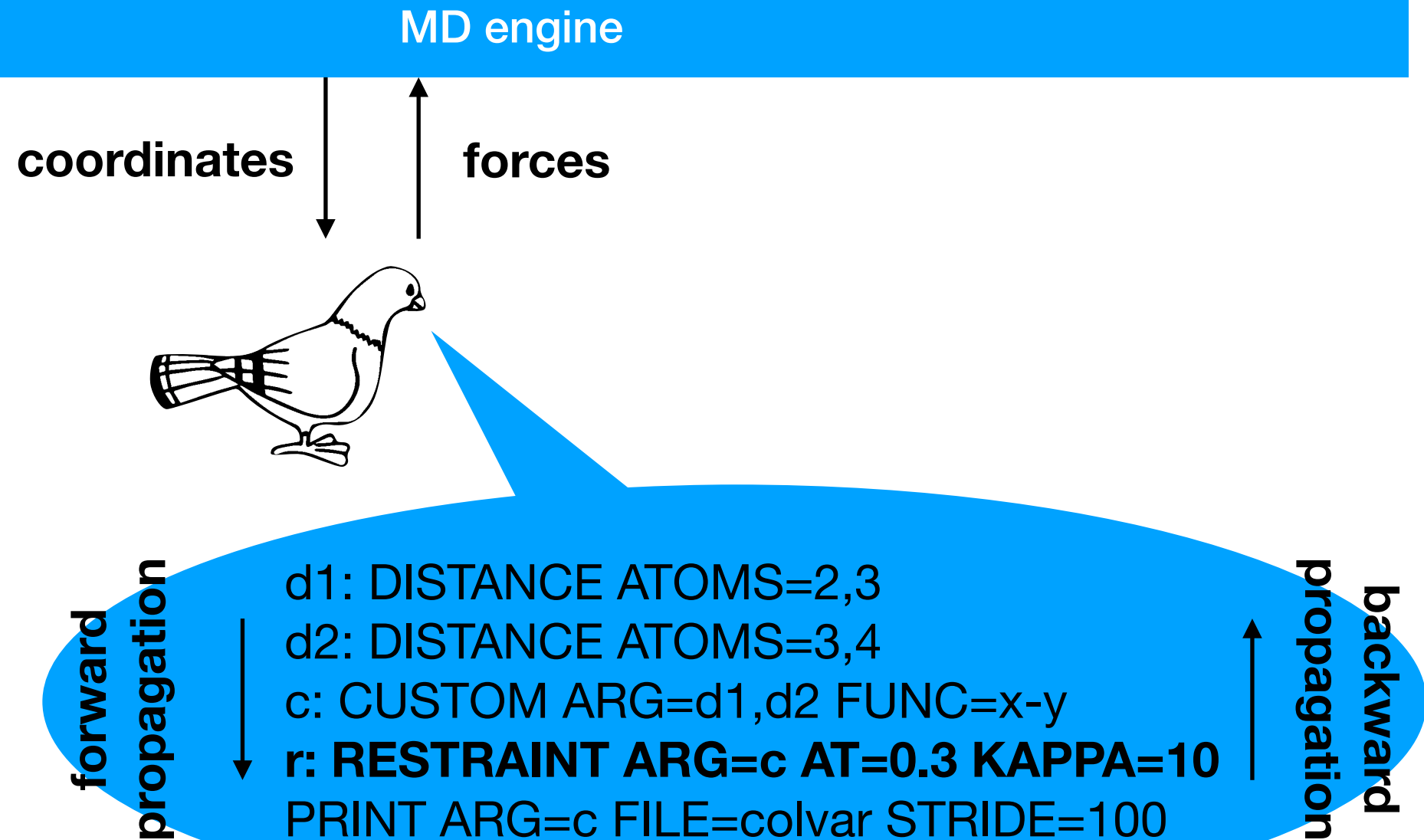
coordinates



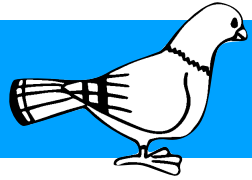
forward
propagation

d1: DISTANCE ATOMS=2,3
d2: DISTANCE ATOMS=3,4
c: CUSTOM ARG=d1,d2 FUNC=x-y
PRINT ARG=c FILE=colvar STRIDE=100

Using PLUMED to bias a simulation

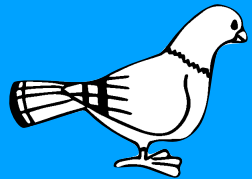


Using PLUMED for WHAM analysis



PLUMED driver on concatenated trajectory

+



REWEIGHT_WHAM action
(integrated with PLUMED)

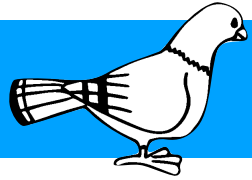
Or



An external python tool, wham.py
(allow more flexibility for bootstrapping)

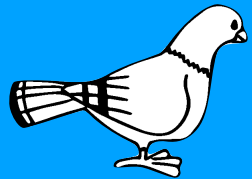
Note: both approaches are bin-less, and allow arbitrary bias potentials to be used in all replicas (i.e., not just a series of restraints on the same variable)

Using PLUMED for WHAM analysis



PLUMED driver on concatenated trajectory

+



REWEIGHT_WHAM action
(integrated with PLUMED)

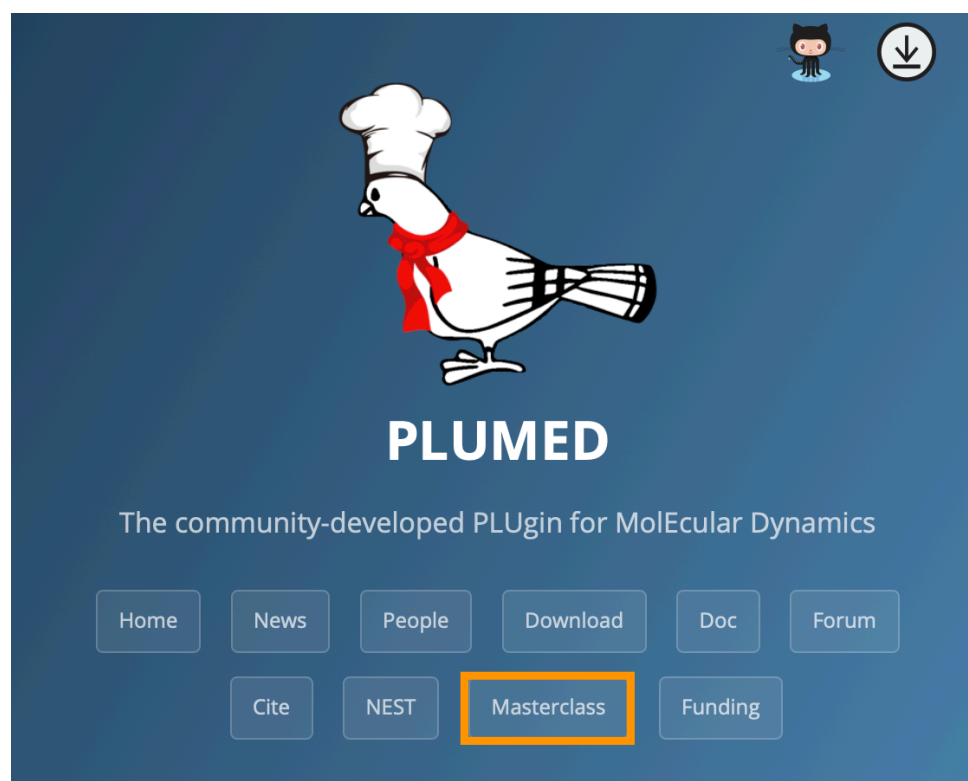
Or



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Note: both approaches are bin-less, and allow arbitrary bias potentials to be used in all replicas (i.e., not just a series of restraints on the same variable)

Instructions



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1. Go to www.plumed.org
2. Click on the **Masterclass** tab
3. Click on the **Topic** of class 21.3
4. 1 week to complete the exercises
5. Questions/discussions on Slack channel [masterclass-21-3](#)
6. Lecture I and II available on **YouTube**